

1. Introduction to JSP (JavaServer Pages)

What is JSP?

JSP (JavaServer Pages) is a technology used to create **dynamic web pages** using Java.

It allows you to embed Java code inside HTML.

Why JSP is used?

- Simplifies web development
 - Separates **presentation (HTML)** from **business logic**
 - Automatically converted into a servlet by the server
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How JSP Works

1. JSP file is requested
 2. Server converts JSP → Servlet
 3. Servlet is compiled and executed
 4. Response is sent to client
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2. Key Components of JSP

A. Scriptlets

What are Scriptlets?

Scriptlets are blocks of **Java code embedded inside JSP**.

Syntax

```
<%  
    // Java code  
%>
```

Example

```
<%  
    int x = 10;  
    out.println(x);  
%>
```

Note

- Used in older JSP
 - Not recommended in modern development (use JSTL instead)
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B. JSTL (JavaServer Pages Standard Tag Library)

What is JSTL?

JSTL is a collection of **predefined tags** used to perform common tasks in JSP.

Why use JSTL?

- Reduces Java code in JSP
 - Cleaner and more readable
 - Easier maintenance
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Common JSTL Tags

Tag	Purpose
<c:out>	Display data
<c:if>	Conditional statement
<c:forEach>	Looping
<c:set>	Set variable

Example

```
<c:if test="${age > 18}">  
  Adult  
</c:if>
```

C. Custom Tags

What are Custom Tags?

Custom tags are **user-defined tags** created to reuse functionality in JSP.

Why use Custom Tags?

- Encapsulate complex logic

- Reusability
 - Cleaner JSP pages
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Example (concept)

```
<mytag:welcome />
```

Defined using:

- Tag handler class
 - TLD (Tag Library Descriptor)
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D. Implicit Objects

What are Implicit Objects?

These are **predefined objects** available in JSP without declaration.

Important Implicit Objects

Object	Description
request	Client request data
response	Response to client
session	Session data
application	Application-wide data
out	Output writer
config	Servlet configuration
pageContext	Page-related data
page	Current JSP page
exception	Exception handling

Example

```
<%  
    String name = request.getParameter("name");  
    out.println(name);  
%>
```